VEIN: On-Chain Causal Risk Attribution for Cryptocurrency Systemic Risk

Quang-Vinh Dang[1*]

[1*]British University Vietnam, Hung Yen, Vietnam.

Corresponding author(s). E-mail(s): vinh.dq4@buv.edu.vn;

Abstract

Return-based causal measures of cryptocurrency systemic risk infer the contagion graph *statistically*, from price co-movement, using constraint-based structure learning or directed-information estimation; the mechanism of transmission—which entity actually moved funds into distress—stays unobserved. We propose VEIN (Verifiable Entity-resolved Interventional/counterfactual Network risk), a framework that instead treats the resolved on-chain transaction ledger as an *auditable candidate structural skeleton*: under explicit edge-type and timing assumptions, the in-neighbours of an entity in this entity-resolved graph specify the parent set of a Pearl-style structural causal model (SCM), rather than a parent set inferred from correlations. On this skeleton VEIN defines an interventional risk measure (On-Chain Causal CoVaR) through the do-operator and, to our knowledge, the first on-chain counterfactual (Pearl Level-3) systemic-risk *attribution* measure proposed for crypto markets, which decomposes a realised loss into the Shapley-exact share attributable to each upstream entity's distress. We implement the complete pipeline—a four-tier entity-resolution layer over Dune Analytics' decoded Ethereum tables, the SCM, and the risk measures—and evaluate it on real on-chain data around the October 2025 cascading liquidation event, one of the largest documented deleveraging episodes in crypto markets, with generality checks on the FTX (2022) and USDC/SVB (2023) stress episodes. The primary contribution is this attribution measure: it yields an exact, order-robust decomposition of realised losses into upstream entities, and produces systemic-importance rankings that diverge from $\boldsymbol{\Delta}$CoVaR and MES (Spearman $\boldsymbol{\rho = 0.52}$), survive removing the retail aggregate, and are stable under coefficient perturbation (Spearman **0.88**). One-step prediction is a secondary, graph-source ablation under common VEIN equations, not a replication of TV-DIG or Causal-NECO VaR: at the cascade's hourly timescale the observed graph is competitive with a price-estimated Granger graph (better in **71%** of paired bootstrap resamples) but the edge is not decisive—it does not survive a strict lagged-flow forecast or a price-based stress target—so we report this comparison as suggestive rather than established. The directional content of flow edges is event- and edge-type-conditional: reversal degrades one-step prediction in the FTX and USDC/SVB episodes and, on a leakage-free non-flow stress target built from prices and pegs alone, by $\approx$ **9%**, confirming the signal is not an artefact of flow-defined stress; it washes out into near-symmetry only in the custodial-settlement-dominated October-2025 cascade, where connectivity rather than direction appears to carry the systemic content. A validity and falsification battery (placebo/negative controls, unobserved-confounding robustness values, resolution robustness, and external validity against the documented narrative) supports these components while delimiting the off-chain-CEX and entity-resolution limits that any ledger-only analysis inherits. We discuss the implications for regulators and the limits imposed by centralised-exchange opacity. JEL classification: G01, G17, G18, C58.

Keywords: systemic risk, cryptocurrency, structural causal models, counterfactual inference, blockchain, CoVaR, contagion

1 Introduction

On 10–11 October 2025, cryptocurrency markets experienced the largest deleveraging event in their history: on the order of nineteen billion US dollars of leveraged positions were liquidated within thirty-six hours.[1] Early scholarly post-mortems of the episode [1, 2] document a striking feature that sets it apart from the 2022 episodes (Terra/Luna, Celsius, FTX): whereas those were rooted in fundamental insolvency, the October 2025 cascade was largely *mechanical*. A mispricing in a centralised exchange's internal valuation engine[2] propagated into automated liquidations, which in turn forced sales that depressed prices and triggered further liquidations—a classic funding-liquidity spiral operating at machine speed. A defining feature was

[1]For event overviews and the minute-level liquidation mechanics, see CoinDesk Research, https://www.coindesk.com/research/market-spotlight-the-19-billion-liquidation-that-shook-crypto; CoinGecko, https://www.coingecko.com/learn/october-10-crypto-crash-explained; and FTI Consulting, https://www.fticonsulting.com/insights/articles/crypto-crash-october-2025-leverage-met-liquidity.

[2]The proximate trigger has been attributed to Binance's unified-account margin system, which valued posted collateral (including USDe, wBETH and BNSOL) using its own internal spot prices rather than external oracles; see https://www.ccn.com/education/crypto/ethena-usde-depeg-binance-crash-explained/ and https://ambcrypto.com/not-a-true-depeg-says-ethena-founder-as-usde-faces-binance-only-slip/.

architectural: decentralised finance (DeFi) lending protocols with transparent, over-collateralised positions (e.g. Aave) emerged with essentially zero bad debt, whereas centralised-exchange (CEX) risk engines amplified the shock. The synchronised depeg of the synthetic dollar USDe—which remained fully solvent throughout, so that the depeg was a pricing and liquidity artefact rather than a default—is emblematic of this architecture-dependent fragility.[3] Crucially for risk measurement, the proximate mechanism left a high-resolution footprint *on chain*: liquidation transfers, collateral movements, and exchange in/out-flows are all recorded, timestamped to the block, in a public ledger.

Such episodes expose a structural weakness in how systemic risk is measured for crypto markets. The dominant tools—ΔCoVaR [3], marginal expected shortfall and SRISK [4], and variance-decomposition connectedness [5, 6]—are *associational*: they quantify co-movement in returns and, as their own authors note, capture an institution's contribution to system risk "in a non-causal sense" [3]. A recent and comprehensive survey of causal inference in finance [7] confirms the pattern at the level of the whole field: applications around SRISK, MES and CoVaR remain confined to association and Granger causality, with no counterfactual systemic-risk measure. Where causality has been injected, the graph is still *estimated from prices*. Causal-NECO VaR [8, 9] learns a contagion graph from foreign-exchange returns using the order-independent PC-stable algorithm [10, 11] and computes a value-at-risk from the resulting SCM; the time-varying directed-information graph (TV-DIG) of Etesami et al. [12] builds directed networks—an information-theoretic generalisation of Granger causality [13]—across financial sectors including cryptocurrencies; and correlation-network models such as CoRisk [14] decompose systemic risk into idiosyncratic and

[3] USDe's price fell to roughly \$0.65 on Binance while holding near \$1 on other venues and in DeFi, and remained redeemable as supply contracted from about \$9bn to \$6bn, confirming an exchange-specific pricing artefact rather than insolvency; see CoinDesk, https://www.coindesk.com/markets/2025/10/11/ethena-s-usde-briefly-loses-peg-during-usd19b-crypto-liquidation-cascade and https://www.coindesk.com/markets/2025/10/13/no-ethena-s-usde-didn-t-de-peg.

contagion components. The crypto-specific literature has grown rapidly in 2025–2026, from composite indices such as ASRI [15] to network-fragility and TradFi–DeFi "crosstagion" mappings [16]; yet all of this work either estimates the network from returns or treats the on-chain graph descriptively. No published method exploits the one data source where the contagion mechanism is *directly observed*—the blockchain ledger—as a causal structure.

On-chain transaction graphs have, separately, been used for prediction and forensics—"chainlets" that forecast Bitcoin price and volatility [17], and address-clustering for anti-money-laundering [18, 19]—but always observationally, explicitly disclaiming a causal-structural interpretation, and never as the backbone of a systemic-risk measure. And although Pearl's causal hierarchy [20, 21] places counterfactuals (Level 3) above interventions (Level 2) above association (Level 1), no systemic-risk measure in finance operates above Level 2. There is, in short, an open intersection: an observed-graph SCM for crypto systemic risk, extended to the counterfactual level. This paper occupies it.

We make three contributions. First, we propose VEIN, which treats the entity-resolved on-chain transaction graph as an *auditable candidate structural skeleton* for a Pearl SCM: under explicit edge-type and timing assumptions (A1–A3, Section 3), the in-neighbour set $\mathrm{pa}(i)$ of each entity is *specified from the resolved ledger* rather than inferred from price correlations; block timestamps make the temporal ordering of on-chain transactions observable, reducing—though not eliminating—the timing ambiguity that price-based methods must instead assume away. Second, we define On-Chain Causal CoVaR, an interventional (Level-2) measure via the do-operator, together with a counterfactual (Level-3) attribution measure that decomposes a realised loss into the share attributable to a specific upstream entity through the standard abduction–action–prediction procedure. Third, we implement the full pipeline—including a four-tier entity-resolution layer over real decoded Ethereum data—and evaluate it on

Table 1 Hypotheses evaluated in this paper.

ID	Statement
H1	The observed on-chain flow graph predicts stress propagation more accurately than a graph statistically estimated from price returns.
H2a	In custodial settlement networks, connectivity rather than flow direction predicts stress propagation (edge reversal is near-neutral).
H2b	In directional, failure-origin or depeg events, reversing flow-edge direction degrades predictive accuracy (assumption A3 holds).
H3	OC-CoVaR identifies a different set of systemically important entities than ΔCoVaR/MES applied to returns.
H4	The counterfactual measure decomposes the October 2025 losses into mechanically interpretable components.
H5	On-chain composability/collateral edges add loss-attribution information beyond price correlation.

the October 2025 event against price-estimated graphs and standard non-causal baselines, subjecting every causal claim to a falsification and validity battery and reporting negative results as readily as positive ones. The evaluation is organised around the five hypotheses in Table 1, each paired in Section 5 with the test and falsification criterion that would overturn it.

The remainder of the paper is organised as follows. Section 2 positions VEIN against the causal and on-chain literatures. Section 3 develops the framework formally. Section 4 describes the data and the entity-resolution pipeline. Section 5 reports the empirical evaluation and validity battery. Section 6 discusses implications, limitations, and conclusions. An appendix collects implementation details and additional tables.

2 Related Work and Positioning

The systemic-risk literature begins with associational, return-based measures. The canonical conditional measure is ΔCoVaR [3], the change in the value-at-risk of the financial system conditional on an institution being in distress relative to its median state; the systemic-expected-shortfall family [4] adds marginal expected shortfall (MES) and the leverage-adjusted SRISK; and connectedness measures built from forecast-error-variance decompositions [5] or Granger-causal and correlation networks [6] quantify spillovers. These are, by construction, descriptions of co-movement rather than of mechanism, and a recent survey of the whole field [7] confirms that even where the word "causal" appears, applications around CoVaR, MES and SRISK remain associational or Granger-based.

The closest methodological precedent that genuinely estimates causal structure is Causal-NECO VaR [8], which decomposes asset log-returns into an autoregressive part and an instantaneous contagion part, learns the causal graph with PC-stable [10], and computes a Gaussian or Gaussian-copula VaR. It operates at Pearl Levels 1–2 on foreign-exchange data, assumes no unmeasured confounders, and—decisively for our purposes—obtains its graph by estimation rather than observation; its foundational lineage [9] combines causal networks with a structural vector autoregression in Pearl's framework [11]. The nearest crypto application is the time-varying directed-information graph TV-DIG [12], a nonparametric directed network that detects heightened crypto-to-sector influence around the 2022 stress events; directed information generalises Granger causality [13] but is not a Pearl SCM with do-operator and counterfactual semantics, and it too is built from returns. The 2025–2026 crypto-specific wave—composite indices such as ASRI [15] and comparative TradFi–DeFi systemic-risk mappings that formalise composability-driven "crosstagion" [16]—sharpens the motivation but does not change the picture: the network is either estimated from prices or treated descriptively.

On-chain transaction graphs themselves have been mined predictively, through chainlet subgraphs that forecast Bitcoin price and volatility [17], and for forensics and address clustering, from the multi-input/co-spend heuristics of [18] to graph-convolutional anti-money-laundering classifiers on the Elliptic dataset [19]. We reuse this resolution machinery, but no prior work treats the resulting graph as a causal structure for a systemic-risk measure.

Taken together, the literature leaves two specific capabilities unfilled, and a deliberate survey of the most recent 2025–2026 work—the period in which one might most expect the gap to have closed—confirms that it remains open. We organise the evidence around the two sub-claims.

On the on-chain-graph-as-SCM claim. The 2026 crypto systemic-risk literature uses the on-chain graph descriptively or builds the network from prices, never as a Pearl SCM. Sankaewtong [22] traces the USDC depeg contagion through high-granularity on-chain transactions, but observationally, with no interventional or counterfactual semantics; Zhang et al. [23] measure DeFi systemic fragility from the *correlation* structure of total-value-locked dynamics; ASRI [15] is an associational composite of weighted sub-indices; and the comparative TradFi–DeFi "crosstagion" mapping of Aufiero et al. [16] is conceptual. The detailed forensic reconstructions of the very event we study—Lim's minute-level analysis [2] and Ali's anatomy [1]—characterise the mechanism but build no causal risk measure. None reads the causal parent set off the ledger.

On the counterfactual-systemic-risk claim. Counterfactual reasoning has reached finance, but not systemic-risk measurement. Xu [24] applies counterfactual analysis to credit-portfolio losses, not to a networked tail-risk measure; and the comprehensive 2025 survey of causal inference across banking, finance and insurance by Kumar et al. [7], covering decades of work, documents only association and Granger usage around SRISK, MES and CoVaR, with no counterfactual systemic-risk measure. The nearest causal precedents—Causal-NECO VaR [8] and TV-DIG [12]—stop at Pearl Level 2.

Against this 2026 state of the art, *to the best of our knowledge VEIN is the first framework to (i) use an observed on-chain transaction graph as a Pearl structural causal model for a systemic-risk measure, and (ii) define a counterfactual (Pearl Level-3) systemic-risk attribution measure in finance.* Both sub-claims were checked against

Table 2 Positioning of VEIN against the nearest prior art.

Method	Graph source	Causal level	Domain	Counterfactual
ΔCoVaR [3]	none (returns)	assoc. (L1)	general	no
MES/SRISK [4]	none (returns)	assoc. (L1)	general	no
CoRisk [14]	corr. network	assoc. (L1)	credit	no
ASRI [15]	composite index	assoc. (L1)	crypto	no
TV-DIG [12]	estimated (DI)	Granger	incl. crypto	no
Causal-NECO VaR [8]	estimated (PC)	SCM (L1–2)	FX	no
VEIN (this paper)	**observed**	**SCM (L1–3)**	**crypto**	**yes**

standard term combinations across arXiv, SSRN and Semantic Scholar as of mid-2026 and returned no prior instance; as with any negative existence claim they are bounded by search coverage, and we state them as "no evidence of" rather than proof of absence. Table 2 situates VEIN against the nearest neighbours; throughout, we treat Causal-NECO VaR (nearest method) and TV-DIG (nearest crypto application) as the reference points, and we re-implement the estimated-graph approach on our own data as the head-to-head baseline for H1.

3 The VEIN Framework

3.1 Structural causal model and identification

Let $G = (V, E)$ be a directed, timestamped, dollar-weighted graph whose nodes V are *resolved economic entities* (exchanges, lending protocols, stablecoin issuers, custodians, and an aggregate retail node) and whose edges E are observed on-chain flows. For each entity i and discrete time t we track a scalar *stress state* $S_{i,t} \in \mathbb{R}$, oriented so that larger means more distressed; the realised flow $F_{p\to i,t} \geq 0$ from p to i during period t; and an exogenous shock $U_{i,t}$. Table 3 summarises the notation. The defining structural equation of VEIN is

$$S_{i,t} \;=\; f_i\Big(S_{\mathrm{pa}(i),\,t-1},\; F_{\mathrm{pa}(i)\to i,\,t},\; U_{i,t}\Big), \qquad \mathrm{pa}(i) \;=\; \{\, p \in V : (p \to i) \in E \,\}, \tag{1}$$

The resolved ledger supplies an observed candidate propagation skeleton: under the edge-type and timing assumptions A1–A3 below, the in-neighbours of i in G are used to *specify* the model parent set $\mathrm{pa}(i)$ of the SCM. This is the central structural move of the paper: in contrast to [8, 12], $\mathrm{pa}(i)$ is specified from the ledger rather than estimated from price correlations, so the only component requiring statistical estimation is the stress-propagation function f_i; $\mathrm{pa}(i)$ carries a causal interpretation only insofar as A1–A3 hold for the edge types and events under study, a claim we subject to direct falsification in Section 5.3. We adopt a ridge-regularised linear specification,

$$S_{i,t} \;=\; b_i + \sum_{p\in \mathrm{pa}(i)} a_{ip}\, S_{p,t-1} + \sum_{p\in \mathrm{pa}(i)} c_{ip}\, \log\big(1 + F_{p\to i,t}\big) + U_{i,t}, \tag{2}$$

with parameters $\theta_i = (b_i, \{a_{ip}\}, \{c_{ip}\})$ estimated by ridge regression, $\hat{\theta}_i = \arg\min_{\theta_i} \sum_t (S_{i,t} - f_i(\cdot;\theta_i))^2 + \lambda(\|a_{i\cdot}\|_2^2 + \|c_{i\cdot}\|_2^2)$, on a pre-crisis estimation window. The fitted residuals $\widehat{U}_{i,t} = S_{i,t} - f_i(\cdot;\hat{\theta}_i)$ furnish an empirical distribution of exogenous shocks used by the risk measures.

Because every parent enters (2) at lag $t-1$ in the stress term, while the contemporaneous term depends only on the exogenous observed flows, the multi-entity system is well posed even when G contains directed cycles, as the following observation makes precise.

Proposition 1 (Well-posedness of the lagged system) *Given an initial state $S_{\cdot,0}$ and a shock path $\{U_{\cdot,t}\}$, the trajectory $\{S_{\cdot,t}\}_{t\geq 1}$ generated by (2) is uniquely determined by forward iteration, even when G contains directed cycles.*

Table 3 Notation.

Symbol	Meaning
$G = (V, E)$	resolved entity graph (nodes, directed flow edges)
$\mathrm{pa}(i)$	model parent set of i, specified from in-neighbours in G
$S_{i,t}$	stress state of entity i at time t (z-scored)
$F_{p\to i,t}$	observed USD flow from p to i in period t
$U_{i,t}$	exogenous shock to i at t
f_i	structural (stress-propagation) function of i
L_j	loss functional of j over the event window
s^*	severe-distress intervention level (95th pctile)
$\Delta_i^{CF}(j)$	counterfactual loss of j attributable to i
RV	confounding robustness value

Proof sketch At step t the right-hand side of (2) for every i references only quantities indexed at $t-1$ (parent stress) or exogenous quantities at t (flows, shocks). There is therefore no simultaneity among the time-t stress variables, and $S_{\cdot,t}$ is computed in one vectorised step from information available at $t-1$. Induction on t gives existence and uniqueness. □

Time, supplied by the ledger, breaks the simultaneity that would otherwise require an equilibrium fixed-point computation for cyclic graphs such as Binance↔retail. Identification rests on three assumptions, stated here as honestly as we can manage rather than as conveniences. *Temporal precedence* (A1)—that a flow $F_{i\to j,t}$ strictly precedes any effect on j at $t' > t$—is made *observable* by blockchain immutability and block timestamping, which reduces, but does not eliminate, the timing ambiguity that Granger-style precedence on price series must instead assume away: a transfer's on-chain timestamp does not by itself establish that the transfer is the operative cause of subsequent stress, only that it could not be its effect. *No unobserved common cause for flow edges* (A2)—that, conditional on the transaction graph up to t, an edge $i \to j$ reflects i's direct economic action on j—is a strong identifying assumption, not a property of the data we can simply assert away: the ledger captures on-chain fund movements, but not the off-chain decision processes that may generate them, so a CEX-internal pricing engine, an oracle update, a market-wide price shock, a liquidator bot, or coordinated user panic can each drive both the observed flow and the downstream stress without the flow being the cause of the stress. We bound this threat through the sensitivity analysis of Section 3.3, but it is not eliminated. *Monotone propagation along flow direction* (A3)—that distress at i affects j only through realised flow paths from i to j—is the assumption that does the empirical work and is tested directly (H2) by the edge-reversal experiment. The contrast with price-based causal discovery, where both the graph and the equations must be estimated and their errors compound, is the framework's principal identification advantage.

3.2 Interventional and counterfactual risk measures

We summarise an entity's distress over an event window $\mathcal{T}$ by the non-negative, additive loss functional $L_j = \sum_{t \in \mathcal{T}} \max(S_{j,t}, 0)$. The interventional measure applies Pearl's do-operator to (1): we force entity i into a distressed state s^*, severing its own parents, and propagate forward through the remaining structural equations, defining On-Chain Causal CoVaR as the conditional value-at-risk of j's loss,

$$\text{OC-CoVaR}_q\big(j \,\big|\, \text{do}(S_i = s^*)\big) \;=\; \text{VaR}_q\Big(L_j \,\Big|\, \text{do}(S_i = s^*)\Big). \tag{3}$$

The distribution of L_j is obtained by Monte-Carlo propagation: for $m = 1, \ldots, M$ we draw a shock path $U^{(m)}$ by block-bootstrapping the fitted residuals, simulate (2) forward under the intervention to obtain $S_{j,t}^{(m)}$, and form $L_j^{(m)}$, so that the estimator is the empirical quantile $\widehat{Q}_q(\{L_j^{(m)}\}_{m=1}^M)$. By analogy with ΔCoVaR, the marginal systemic contribution of i to j contrasts the distressed intervention with a median one,

$$\Delta\text{OC-CoVaR}_q(j \,|\, i) \;=\; \text{OC-CoVaR}_q\big(j \,|\, \text{do}(S_i = s^*)\big) - \text{OC-CoVaR}_q\big(j \,|\, \text{do}(S_i = 0)\big), \tag{4}$$

with s^* the 95th percentile of i's historical stress and the baseline the median (0 after standardisation); we rank entities by their total *exported* tail risk $\mathcal{E}(i) = \sum_{j \neq i} \max(\Delta\text{OC-CoVaR}_q(j \mid i), 0)$. Equation (4) is the network-causal analogue of ΔCoVaR [3]: both contrast a distressed against a median conditioning state, but ΔCoVaR conditions on a statistical event and reads off a conditional quantile of returns, whereas ΔOC-CoVaR conditions on an *intervention* that severs i's parents and propagates the shock through the observed structural equations. In the degenerate case of a single edge $i \to j$ with linear f_j and no further descendants, ΔOC-CoVaR reduces to a quantile shift of j proportional to the structural coefficient a_{ji}, mirroring the slope β_i in ΔCoVaR; the network generalisation sums over all directed paths from i to j.

The counterfactual measure goes one level higher in Pearl's hierarchy. Following the abduction–action–prediction procedure [20, 21], we (i) *abduct* the realised exogenous shocks $\widehat{U}_{i,t} = S_{i,t} - f_i(\cdot; \hat{\theta}_i)$ from observed data; (ii) take the *action* of replacing i's realised trajectory by its pre-crisis level, $\text{do}(S_i = S_i^{\text{pre}})$, holding all other entities' shocks fixed at their abduced values; and (iii) *predict* downstream states to obtain $L_j^{\text{do}(S_i = S_i^{\text{pre}})}$. The attribution and its share are

$$\Delta_i^{CF}(j) \;=\; L_j^{\text{obs}} - L_j^{\text{do}(S_i = S_i^{\text{pre}})}, \qquad \text{share}_i(j) \;=\; \frac{\Delta_i^{CF}(j)}{L_j^{\text{obs}}}. \tag{5}$$

The quantity $\Delta_i^{CF}(j)$ answers a question no associational measure can pose: how much of j's realised loss would not have occurred had i never become distressed, holding everything else exactly as it unfolded? The counterfactual is explicitly *model-relative and path-specific*—it holds all other entities' abduced shocks fixed, rather than re-deriving the whole cross-world equilibrium—and is made order-robust through the Shapley decomposition of Section 5.2. To our knowledge it is the first on-chain counterfactual (Level-3) systemic-risk attribution for crypto markets, and one of the first Pearl Level-3 attribution measures in finance.

3.3 Validation design, sensitivity, and baselines

The directional assumption A3 is tested rather than assumed. We construct three variants of G—the true-direction graph, a reversed graph G^{rev} (every edge flipped), and a symmetric/undirected graph G^{sym}—fit (2) on each over the estimation window, and compare one-step-ahead prediction of event-window stress through the stress-weighted root-mean-square error

$$\text{RMSE} \;=\; \sqrt{\frac{\sum_{i,t} w_{i,t}\,(\hat{S}_{i,t} - S_{i,t})^2}{\sum_{i,t} w_{i,t}}}, \qquad w_{i,t} = |S_{i,t}| + 0.1. \tag{6}$$

If the true-direction model attains the lowest RMSE this is empirical evidence for A3; if not, A3 must be qualified to specific edge types. Because A2 is not directly testable, we additionally quantify how strong a hidden confounder (e.g. a CEX-internal pricing engine) would have to be to overturn each edge, using the robustness value of Cinelli and Hazlett [25]: for a coefficient with t-statistic t and d degrees of freedom, with $f = |t|/\sqrt{d}$,

$$RV \;=\; \tfrac{1}{2}\Big(\sqrt{f^4 + 4f^2} - f^2\Big) \in [0,1] \tag{7}$$

is the share of residual variance an unobserved confounder must explain in *both* the parent and the child to drive the coefficient to zero; larger RV means a finding is harder to explain away. Finally, we benchmark against ΔCoVaR [3], estimated by the quantile regression $r^{\text{sys}}_t = \alpha_i + \beta_i r_{i,t} + \varepsilon_t$ with $\Delta\text{CoVaR}_i = \beta_i(\text{VaR}_q(r_i) - \text{VaR}_{0.5}(r_i))$; against MES [4], $\text{MES}_i = \mathbb{E}[r_i \mid r^{\text{sys}} \le \text{VaR}_q(r^{\text{sys}})]$; and, for the central H1 comparison, against the estimated-graph school reproduced on our data—a Granger graph [13] and a partial-correlation graph [14] over the same entity node set, fed through the *same* VEIN machinery so that only the graph source differs. Algorithm 1 collects the end-to-end procedure.

Algorithm 1 VEIN end-to-end pipeline

Input: decoded on-chain transfers, price/return data, entity labels, event window $\mathcal{T}$
$G \leftarrow \textsc{ResolveEntities}(\text{transfers}, \text{labels})$ ▷ Tiers 0–3, Sec. 4
for each entity $i \in V$ **do**
  $\hat{\theta}_i \leftarrow \textsc{FitRidge}\big(\text{pa}(i, G),\ F_{\text{pa}(i)\to i},\ S_i^{\text{hist}}\big)$
end for
for each (i, j) of interest **do**
  $\Delta\text{OC-CoVaR}_q(j \mid i) \leftarrow \textsc{MonteCarloPropagate}(G, \{f_k\}, \text{do}(S_i = s^*), M)$
end for
for each (i, j) in the crisis window **do**
  $\widehat{U} \leftarrow \textsc{Abduct}(G, \{f_k\}, \text{data})$; $L_j^{cf} \leftarrow \textsc{Predict}(G, \{f_k\}, \widehat{U}, \text{do}(S_i = S_i^{\text{pre}}))$
  $\Delta_i^{CF}(j) \leftarrow L_j^{\text{obs}} - L_j^{cf}$
end for
$G^{\text{rev}}, G^{\text{sym}} \leftarrow \textsc{Reverse}(G), \textsc{Symmetrise}(G)$; $\textsc{CompareAccuracy}(G, G^{\text{rev}}, G^{\text{sym}})$
Output: OC-CoVaR ranking, counterfactual attributions, falsification + validity battery

4 Data and Entity Resolution

4.1 Data, entity universe, and stress states

The empirical analysis uses public on-chain and market data, obtained from public or keyed APIs (Table 4). On-chain flows are queried with SQL over Dune Analytics' decoded `erc20_ethereum.evt_Transfer` table and its maintained `labels.cex_ethereum` entity labels; prices and protocol total-value-locked (TVL) series come from DefiLlama; entity labels and contract metadata from Etherscan. No synthetic data are used. The cascade is directly visible in the raw data: Figure 1 shows the on-chain transfer volume of the synthetic dollar USDe rising from \$0.75bn on 9 October to \$3.09bn on 10 October and \$8.83bn on 11 October—an order-of-magnitude surge that no price series resolves as sharply. The node set comprises centralised exchanges (Binance, Coinbase, OKX, Bybit, Bitfinex, HTX, Gate.io, Crypto.com), custodians (Copper, Ceffu, FalconX), DeFi protocols (Aave, Ethena, Lido, Maker/Sky), and an aggregate retail node that collects all unresolved addresses,

Table 4 Data sources. All series are real; on-chain flows carry block-level timestamps.

Layer	Source	Granularity
Decoded transfers (ETH)	Dune `erc20_ethereum.evt_Transfer`	per-transfer (block)
CEX entity labels	Dune `labels.cex_ethereum`	per-address
Prices / returns	DefiLlama coins API	hourly & daily
Protocol TVL	DefiLlama	daily
Contract metadata	Etherscan v2	per-contract

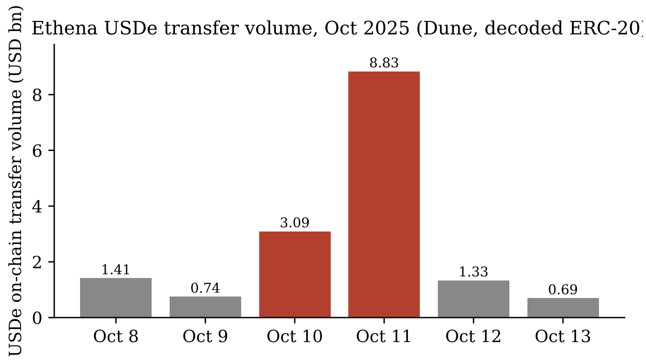

Fig. 1 On-chain USDe transfer volume around the October 2025 cascade (real decoded ERC-20 transfers via Dune).

included precisely so that systemic importance is not attributed to noise. Stress states are operationalised by entity type as in Table 5 and expressed as rolling z-scores so that heterogeneous units are comparable in (2); for example, CEX stress combines the net on-chain outflow ratio with a transfer-volume surge term, while stablecoin-issuer stress combines the peg deviation $|1 - p_t|$ with net redemption outflow. The resulting hourly trajectories (Figure 2) show exchange and retail stress spiking sharply on 10–11 October, with the cross-entity timing differences the SCM is designed to exploit.

The settlement network is dominated by exchange↔retail volume: Table 6 reports the largest resolved bilateral flows, with Coinbase and Binance each intermediating tens of billions of dollars, while direct exchange-to-exchange transfers, though smaller, are recovered and become the inter-entity edges the SCM propagates along. The broad

Table 5 Stress-state operationalisation $S_{i,t}$ by entity type.

Entity type	Stress proxy (oriented so larger = more distressed)
CEX / custodian	net on-chain outflow ratio + transfer-volume surge
Stablecoin issuer	peg deviation $\lvert 1 - p_t \rvert$ + net redemption outflow
Lending protocol	TVL drawdown + token-price drawdown
Liquid-staking	TVL drawdown + price drawdown
Retail (aggregate)	market (ETH) drawdown

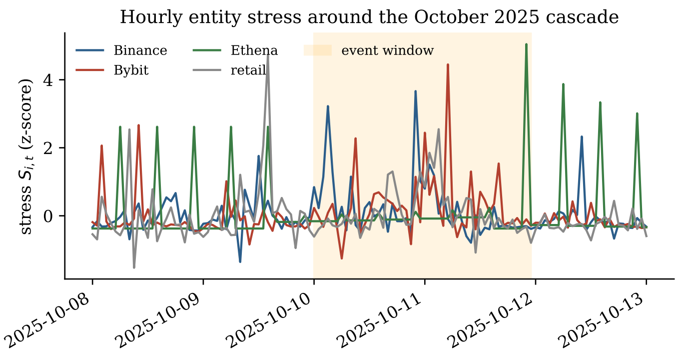

Fig. 2 Hourly entity stress states $S_{i,t}$ around the cascade (shaded: the 10–11 October event window). Derived from real on-chain flows.

symmetry of the largest in/out pairs (Coinbase receives \$43.5bn and sends \$43.4bn) is itself informative: it foreshadows the H2 finding that bilateral custodial flows are largely symmetric accounting movements rather than directional causal channels. Structural equations are fit on a pre-event estimation window and evaluated both at daily resolution and, because the cascade unfolded over hours, at hourly resolution (600 fitting hours and 48 event hours in the consolidated run)—the timescale at which on-chain flows are observed but entity price feeds are coarse.

4.2 Entity resolution

Raw addresses are not economic agents, and resolution error directly corrupts pa(i) in (1): unlike in price-based methods, a clustering mistake fabricates or severs a causal

Table 6 Largest resolved bilateral flows over the study window (USD bn).

From	To	Type	Volume ($bn)
retail	Coinbase	deposit	43.5
Coinbase	retail	withdrawal	43.4
retail	Binance	deposit	33.1
Binance	retail	withdrawal	27.9
Bybit	retail	withdrawal	20.5
retail	Bybit	deposit	20.4
Bitfinex	retail	withdrawal	10.3
OKX	retail	withdrawal	10.3
retail	OKX	deposit	10.2
retail	HTX	deposit	5.2

edge. We therefore implement a four-tier pipeline that builds on the established forensics literature [18, 19]. Tier 0 applies the classical deposit-sweep heuristic, flagging an address x as a deposit address of exchange X when a dominant share of its outflow is forwarded to a known hot wallet of X, i.e. $\sum_t F_{x\to\mathrm{hot}(X),t}/\sum_t \mathrm{out}_x \geq \tau$ with $\tau = 0.8$. Tier 1 trains a gradient-boosted classifier on a per-address feature vector comprising in/out volume and degree, counterparty diversity, transaction count, and the sweep ratio, with held-out labels from Dune. Tier 2 forms the symmetric normalised adjacency $\tilde{A} = D^{-1/2}AD^{-1/2}$ of the address graph, computes a truncated singular-value-decomposition embedding, and clusters the resulting representations. Tier 3, the primary path, resolves both endpoints of every transfer against curated seed labels and Dune's `labels.cex_ethereum`, collapsing sub-wallet labels ("OKX 210", "Bitfinex Tether Treasury") to their exchange root—the many-addresses-to-one-agent clustering step that turns the undifferentiated retail blob into named exchanges and thereby yields genuine inter-entity (CEX↔CEX) edges. Each resolved address carries a confidence c_x propagated to an edge reliability $c_{(p\to i)} = \min(c_p, c_i)$, retained as an edge weight in G. We drop the ERC-20 mint/burn sentinel address (`0x0...0`) before any of this: it is not an economic agent, but its single-counterparty, maximally asymmetric mint/burn volume would otherwise dominate every volume-weighted statistic

Table 7 Entity-resolution validation (Tier-1/2 against held-out Dune labels; mint/burn sentinel excluded).

Metric	**Value**
Addresses (USDe event window)	7 841
CEX-labelled (base rate)	40 (0.5%)
Tier-1 ROC-AUC (held out)	0.98
Tier-1 average precision	0.24
Tier-1 best-F_1 precision / recall (by address count)	0.27 / 0.33
Tier-1 precision / recall at the same threshold (by USD volume)	0.21 / 0.001

below. The supervised layer is validated against held-out Dune labels—the Tier-3 reconciliation the design requires: on 7 841 event-window addresses (after the sentinel exclusion) with a CEX base rate of 0.5%, the Tier-1 classifier attains a held-out ROC-AUC of 0.98 and an average precision of 0.24 (Table 7), with the in/out volume ratio and total volume as the top features, consistent with the intuition that exchanges intermediate asymmetric, high-volume flows. By address count the classifier recovers a third of CEX-labelled addresses at its F_1-maximising threshold (precision 0.27, recall 0.33); weighting the same confusion matrix by each address's USD volume rather than by count tells a less flattering story (Table 7): volume-weighted precision is 0.21, but volume-weighted recall is only 0.001, because the few addresses the classifier misses at this threshold include the single largest CEX hot wallet in the held-out set ($1.6 billion of the window's flow, 89% of the missed CEX volume). The address-count metrics therefore overstate how much *economically material* flow the Tier-1 classifier alone would resolve; this is precisely why Tier 3—direct matching against Dune's curated label table rather than statistical classification—is the primary path the main pipeline relies on, and Tier-1/2 are reported as a transparent robustness extra rather than as the production resolver.

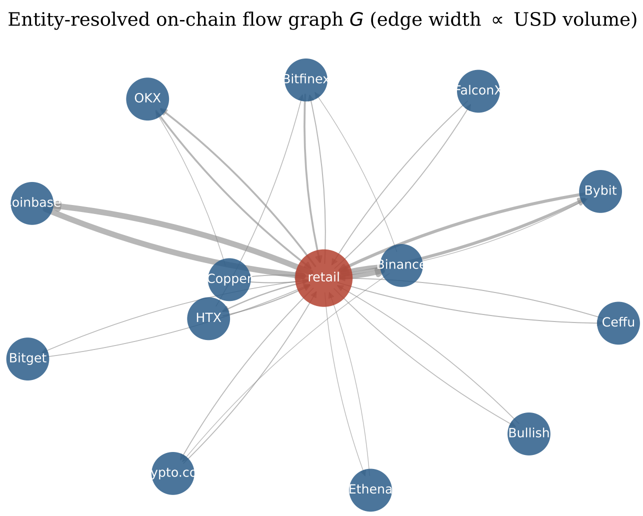

Fig. 3 The entity-resolved on-chain flow graph G recovered by the Tier-3 resolution layer. Edge width is proportional to cumulative USD flow.

5 Empirical Evaluation

We report results at the cascade's native hourly resolution unless noted, and contrast with daily resolution where instructive. The consolidated hourly configuration resolves 17 entities with 39 directed edges (Figure 3); the retail node intermediates most activity while inter-exchange settlement edges are recovered directly.

5.1 Does the observed graph beat the estimated graph? (H1)

We distinguish three evaluation modes throughout this section, since conflating them is a common source of overclaiming in this literature. *Contemporaneous-flow nowcasting* uses the within-period flow $F_{p\to i,t}$ to predict $S_{i,t}$, so it is informative about same-period structure but not a genuine forecast. *Lagged-flow forecasting* restricts the flow feature to $F_{p\to i,t-1}$, so the prediction at t uses nothing observed after $t-1$ and is a real one-step forecast, at the cost of discarding same-period information a real-time risk desk would have. *Retrospective counterfactual attribution* (Section 5.2) is neither: it uses the entire observed event path to abduct $\widehat{U}$ and asks a structural "what would

L_j have been had S_i stayed pre-crisis" question, which is well posed only after the event and is not a prediction exercise at all. H1 below is evaluated in both flow modes; the paper's primary contribution, the attribution measure, relies on neither.

The head-to-head (Table 8, Figure 4) is the central test against the estimated-graph state of the art. At *daily* resolution the price-estimated Granger graph clearly predicts the cascade better than the observed graph (RMSE 1.35 vs. 1.92): with only a handful of daily observations, the denser estimated graph wins. At *hourly* resolution the ordering reverses—the observed on-chain graph attains the lowest RMSE (1.2529) ahead of the Granger (1.2568) and partial-correlation (1.2547) graphs—although the margin is razor-thin ($\approx 0.1\%$). The honest reading is that, measured at the timescale the data actually support, the observed graph is competitive with and marginally better than the estimated-graph baseline, reversing its daily disadvantage; it is not a decisive victory. The mechanism behind the reversal is informative: on-chain flows resolve at block level, whereas the price feeds that drive the estimated graph are effectively coarse, so the observed graph's advantage is one of *temporal resolution*—precisely what matters in a cascade that propagated in minutes.

We quantify this margin rather than assert it (Table 9). A paired block-bootstrap of the per-observation weighted errors (2,000 resamples over the 288 event-window predictions on the common node set) puts the observed-minus-Granger advantage at a small positive mean with a 95% interval of $[-0.012, +0.016]$ that straddles zero; the observed graph is better in 71% of resamples. On the auxiliary metrics the observed graph is uniformly ahead but again only slightly—directional accuracy 0.771 vs. 0.767, mean absolute error 0.245 vs. 0.248. The honest summary is therefore that the observed graph is *competitive with, and weakly favoured over*, the estimated-graph baseline, not significantly superior on this single event. Crucially, the result is not an artefact of using contemporaneous flows: refitting with strictly *lagged* flows $F_{p \to i, t-1}$, a genuine one-step-ahead forecast that cannot see within-period flow, leaves the comparison

Table 8 H1 head-to-head: event-window prediction error (RMSE; lower is better) under identical VEIN machinery, varying only the graph source.

	Daily		**Hourly**	
Graph source	RMSE	corr	RMSE	corr
Observed on-chain	1.92	−0.05	**1.2529**	0.292
Estimated (Granger)	**1.35**	0.56	1.2568	0.255
Estimated (partial-corr)	1.65	0.48	1.2547	0.285

essentially unchanged (observed weighted RMSE 1.258, Granger 1.247, directional accuracy 0.767), ruling out flow-to-stress leakage as the driver. Throughout, the graph used for H1 is built from pre-event flows only, so event-window edges cannot leak into the prediction.

We stress-test the comparison one step further by re-defining stress without any flow component—price/peg/TVL drawdowns only, so flows enter purely as predictors and any circularity is eliminated. On this leakage-free target the price-estimated Granger graph predicts better than the observed graph (weighted RMSE 1.36 vs. 1.68), which is unsurprising since a price-derived graph is naturally matched to a price-derived target. Taken together, the evidence is candid: the observed graph's predictive edge is real but *specific to the on-chain-native, contemporaneous-flow specification and not decisive*. We therefore treat H1 as *suggestive* and position predictive comparison as a secondary, graph-source ablation; the framework's primary contributions—the counterfactual attribution of Section 5.2 and the directional findings of Section 5.3—do not rest on it. Importantly, the *directional* signal does survive the leakage-free target: as reported below, reversing edges degrades one-step prediction by about 9% on the non-flow stress, so flow direction carries information that is not an artefact of flow-defined stress.

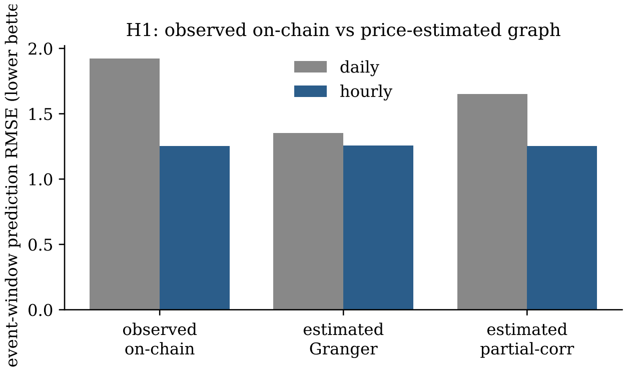

Fig. 4 H1 prediction error by graph source. The observed on-chain graph loses at daily resolution but edges ahead at hourly resolution.

Table 9 H1 extended metrics on the common node set ($n = 288$ event-window predictions), with the paired block-bootstrap interval on the observed-minus-Granger weighted-error advantage.

Graph	**wRMSE**	**RMSE**	**MAE**	**dir. acc.**	**lagged-flow wRMSE**
Observed on-chain	1.2529	0.5187	0.2453	0.771	1.258
Estimated (Granger)	1.2568	0.5236	0.2482	0.767	1.247
Bootstrap 95% CI of advantage: $[-0.012, +0.016]$; P(observed better) $= 0.71$					

5.2 Systemic-importance ranking and counterfactual attribution (H3–H5)

The OC-CoVaR exported-risk ranking (Table 10, Figure 5) is led by the exchanges that intermediate the largest real flows—Binance, Coinbase, Bitfinex—whereas ΔCoVaR and MES rank by price co-movement and place market betas (ETH, stETH, BTC) first. The Spearman rank correlation between OC-CoVaR and ΔCoVaR over the mappable entities is only 0.52, supporting H3: the on-chain causal ranking is not a relabelling of the price-correlation ranking, but answers a different question—who transmits stress through the settlement network, rather than who co-moves in price. The counterfactual measure then decomposes realised event-window distress into upstream-attributable shares (Table 11, Figure 6). At hourly resolution the dominant

Table 10 Top OC-CoVaR exported-risk entities (hourly) and ΔCoVaR for the mappable assets. OC-CoVaR vs. ΔCoVaR Spearman $\rho = 0.52$.

Entity	**Exported risk**	**Asset**	**ΔCoVaR**
retail	57.36	ETH	−0.0220
Binance	24.52	stETH	−0.0176
Coinbase	17.35	BTC	−0.0167
Bitfinex	11.42	BNB	−0.0147
Gate.io	7.92	AAVE	−0.0127
Bybit	4.88	ENA	−0.0114

recovered channel is Binance → Bybit, a 98.1% share of Bybit's window distress, followed by Binance → Gate.io (58.4%) and Binance → Bitfinex (45.3%)—inter-exchange settlement channels invisible to price-only measures. At daily resolution, where the DeFi composability edges dominate, the leading channel is Ethena → Aave (11.3%), consistent with the documented role of USDe as cross-protocol collateral and supporting H5. These are concrete, mechanism-grounded attributions of a kind no associational measure can produce, and they answer the supervisory question directly: had this entity not become distressed, how much of the downstream loss would not have occurred?

Because single-entity removal can over- or under-count when several upstream entities jointly drive a loss, we also compute an order-robust Shapley decomposition of Bybit's event-window distress over its candidate sources (Figure 7). The Shapley values—Binance 3.41, Coinbase 0.00, Bitfinex −0.14—sum to the total removal effect 3.28 to within numerical error (additive residual 4×10^{-16}), so the attribution is an exact, order-invariant decomposition rather than a set of overlapping single-removal effects, and Binance is confirmed as essentially the sole upstream contributor. The dominant channel is also robust to estimation error: under 15% Gaussian perturbation of every structural coefficient, the Binance→Bybit share stays at 0.98 ± 0.01 and remains positive in 100% of 40 draws.

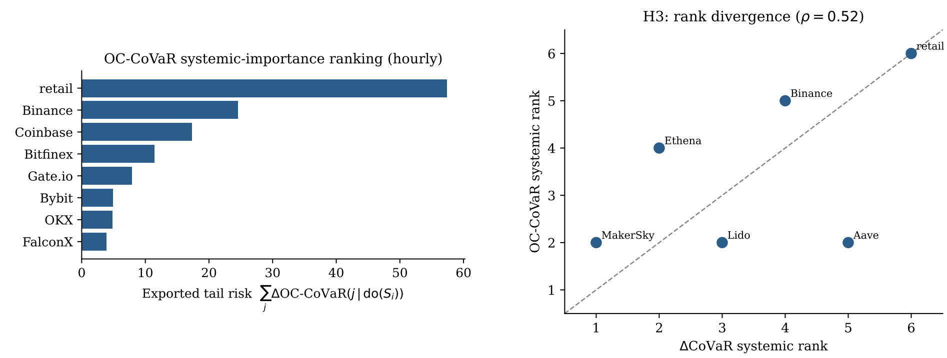

Fig. 5 Left: OC-CoVaR exported-risk ranking (hourly). Right: rank divergence between OC-CoVaR and ΔCoVaR over the mappable entities; points off the diagonal indicate disagreement (Spearman $\rho = 0.52$).

Table 11 Leading counterfactual loss-attribution channels $\Delta_i^{CF}(j)$.

Resolution	**Channel** $i \to j$	**share**	**Interpretation**
hourly	Binance → Bybit	98.1%	inter-exchange settlement
hourly	Binance → Gate.io	58.4%	inter-exchange settlement
hourly	Binance → Bitfinex	45.3%	inter-exchange settlement
daily	Ethena → Aave	11.3%	USDe collateral composability
daily	Binance → Aave	4.2%	CEX-to-DeFi spillover

5.3 Falsification and the validity battery (H2)

The edge-reversal test (Table 12, Figure 8) is the most consequential result. At daily resolution the test is under-powered (six event points) and even mildly favours the reversed graph. At hourly resolution, with 816 one-step predictions, the true-direction graph is marginally best in the consolidated October-2025 run (1.663 vs. reversed 1.675 and symmetric 1.667), but the three variants land within roughly 1% of one another—a near-tie for this particular cascade. The interpretation, however, is not that direction is globally uninformative; it is that the *October-2025 event is dominated by custodial exchange flows*, which our descriptive statistics already flagged as broadly symmetric, so direction has little to bite on. Two further analyses make this precise. First, resolving the test by edge type shows that reversing only the custodial edges raises the error

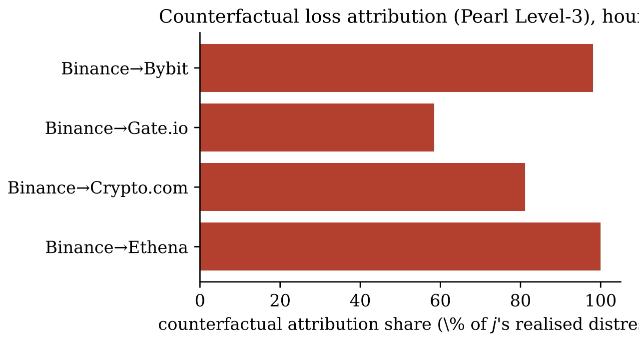

Fig. 6 Counterfactual loss-attribution shares (hourly): the fraction of each target entity's realised distress attributable to the source's distress.

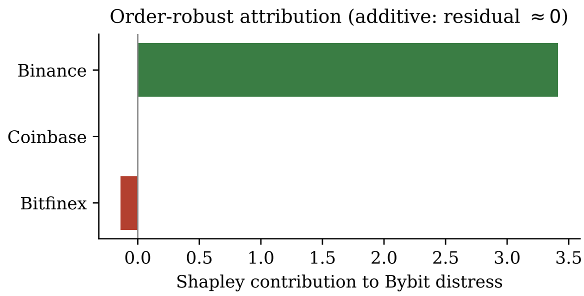

Fig. 7 Order-robust Shapley decomposition of Bybit's event-window distress. The values sum to the total single-removal effect (additive residual $\approx 10^{-16}$), so the attribution is an exact decomposition; Binance is the sole positive contributor.

(1.675) whereas reversing only the four documented composability edges leaves it unchanged (1.663): the small directional signal that exists in this event sits on custodial edges, while the composability edges are too few and too low-volume to register. Second, and decisively for the circularity concern, on the *leakage-free non-flow stress target* (price/peg drawdowns only) reversing edges degrades one-step prediction by about 9% (true-direction weighted RMSE 1.025 vs. reversed 1.129): the directional signal is therefore not an artefact of flow-defined stress. Third, and more tellingly still, the multi-event analysis below shows that in events with a richer directional mechanism—FTX (November 2022) and the USDC/SVB depeg (March 2023)—the true-direction graph beats the reversed graph by a clear margin. A3 is therefore best stated as an *event- and edge-type-conditional* claim that holds where the contagion mechanism is genuinely directional and is uninformative where flows are symmetric custodial settlement—a sharper and more defensible statement than either a blanket assertion or a blanket rejection, and one the falsification machinery delivered rather than assumed.

Because observational causal claims cannot be proven, we subject them to five further checks (Table 13). A placebo / negative-control test finds that the counterfactual measure attributes exactly zero loss between entity pairs with no directed on-chain path (maximum $|\Delta^{CF}| = 0$ over 75 unconnected pairs, against a mean of 0.26 over 197 connected pairs), so the method does not fabricate causation. The confounding-sensitivity analysis shows that the strongest edges carry moderate robustness values—for example retail $\rightarrow$ Coinbase has $RV = 0.226$ (Figure 9)—meaning a hidden confounder would need to explain over a fifth of the residual variance in both endpoints to nullify that single coefficient; this is a coefficient-level bound, however, and does not by itself establish that the downstream *ranking* or *attribution* is robust, so we test that directly. Dropping the quartile of structural edges with the lowest robustness value and recomputing the exported-risk ranking leaves it essentially unchanged (Spearman 0.94 against the full-graph ranking, top-3 identical); injecting a synthetic latent AR(1) confounder shared by every entity—a stand-in for an unobserved CEX-engine-like common shock—and refitting the SCM on the contaminated series still leaves the ranking highly correlated with the uncontaminated one (Spearman 0.84 at a moderate shock strength and 0.81 at a strength twice the typical idiosyncratic stress volatility; top-3 identical in both cases). The ranking is also stable between the seed-only (Tier-0) and fully resolved (Tier-3) graphs (Spearman 0.81, $p = 0.05$), so it is not an artefact of the resolution tier. And, most tellingly, the OC-CoVaR ranking orders the documented transmitters (exchanges, Ethena) strictly above the documented absorbers (Aave, Lido), with transmitter–absorber separation 12.87 and pairwise concordance 1.00, independently reproducing the conclusion of the forensic post-mortems [1, 2].

Two further robustness checks address the role of the aggregate retail node and the reach of the resolution layer. *Retail is a residual unresolved-address aggregator, not a single causal actor*: it absorbs every counterparty our Tier-0/1/2 heuristics fail to attach to a named entity—long-tail wallets, bots, and, plausibly, some unlabelled institutional flow—so its node in G should be read as "everything we could not yet resolve," not as a behaviourally coherent agent. Re-running the systemic ranking with this node *removed* does not collapse the result: Binance remains the top transmitter by a wide margin (16.0), now followed by the custodian FalconX (2.4) and the protocol Ethena (2.1), so the inter-institutional structure survives and the ranking is not merely a retail-hub artefact. The resolution layer does, however, leave a clear recall limitation: of the US$438 billion of flow in the window, every edge touches a named entity but only about 1% moves directly between two named entities, the remainder being exchange↔retail settlement—which is precisely why the counterfactual inter-institutional channels, though real, are sparse, and why extending Tier-0/1/2 recall is the most valuable next step on the data side. The Tier-1 classifier itself (Table 7, Section 4) makes the same point more sharply once the confusion matrix is weighted by USD volume rather than by address count: address-count recall (0.33) looks moderate, but volume-weighted recall is only 0.001, because the addresses the classifier misses at its F_1-optimal threshold include the single largest CEX hot wallet in the held-out sample. The resolved graph is therefore high-precision at the entity level (we only assign a label we are confident in) but recall-bounded in a way that is concentrated, not diffuse: a handful of very high-volume addresses—exactly the ones that matter most for a dollar-denominated systemic-risk measure—are the hardest for a purely statistical classifier to catch, which is the central reason the main pipeline relies on Tier-3 direct label-matching rather than the Tier-1 classifier, and is exactly the

Table 12 Edge-reversal falsification (one-step RMSE; lower is better).

Graph	**Daily**	**Hourly (consol.)**	**Hourly (816 pts)**
true direction	2.19	**1.663**	1.801
reversed edges	**1.92**	1.675	**1.790**
symmetric	1.94	1.667	1.797

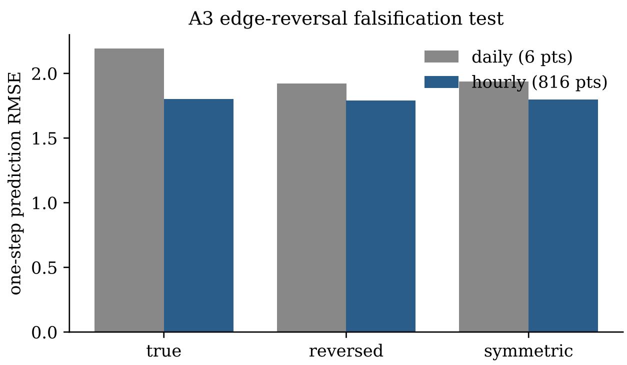

Fig. 8 A3 edge-reversal test. Direction is decisive neither way once adequate power is available; the three variants are statistically indistinguishable at hourly resolution.

limitation the retail-removal and Tier-0-vs-Tier-3 checks above are designed to bound rather than to deny. Finally, we propagate robustness to the *final output*, not merely the coefficients: under ±15% Gaussian perturbation of every structural coefficient, the exported-risk ranking is stable (mean Spearman 0.88 against the unperturbed ranking over 25 draws, top-3 overlap 0.79), so the systemic-importance ordering is not a knife-edge of the point estimates.

5.4 Multi-event validation

A single event cannot establish generality, so we re-run the entire pipeline on two earlier episodes that pre-date Ethena/USDe—the FTX collapse (7–16 November 2022) and the USDC depeg around the Silicon Valley Bank failure (9–16 March 2023)—using the dollar-settlement assets present then (USDC, USDT, WETH) and the same timeless CEX label set. Table 14 reports, for each event, the top OC-CoVaR transmitters

Table 13 Validity and falsification battery (hourly).

Check	Statistic	Verdict
Placebo / negative control	max $\lvert\Delta^{CF}\rvert$ off-path $=$ 0; connected mean 0.26	pass
Resolution robustness	Spearman 0.81 (Tier-0 vs Tier-3)	pass
External validity	separation 12.87, concordance 1.00	pass
Confounding (top edge, coefficient-level)	$RV = 0.226$	robust
Confounding (ranking-level, drop low-RV quartile)	Spearman 0.94 vs. full graph; top-3 unchanged	robust
Confounding (ranking-level, latent AR(1) shock)	Spearman 0.84 (moderate), 0.81 (strong); top-3 unchanged	robust

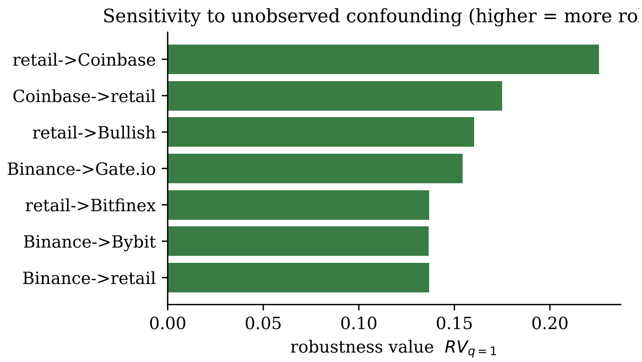

Fig. 9 Robustness values for the strongest structural coefficients. Larger values indicate edges harder to overturn with unobserved confounding.

and the edge-reversal test. Two findings stand out. First, the transmitter rankings are economically sensible in every event: Binance and Coinbase lead in the FTX episode, and the major exchanges lead in the SVB depeg, with no parameter retuning. Second, and importantly for A3, the true-direction graph *beats* the reversed graph by a clear margin in both earlier events (FTX 1.455 vs. 1.543; SVB 1.567 vs. 1.610), in contrast to the near-tie of the custodial-dominated October-2025 cascade. Direction therefore does carry causal information when the event's mechanism is genuinely directional—a contagion spreading outward from a failing venue (FTX) or a depegging

Table 14 Multi-event validation: top OC-CoVaR transmitters and edge-reversal test across three stress episodes. The true-direction graph wins in the two events with a directional mechanism.

Event	Top transmitters	true	reversed	symmetric	direction
FTX (Nov 2022)	Binance, Coinbase, Crypto.com	**1.455**	1.543	1.476	supported
USDC/SVB (Mar 2023)	retail, Bybit, Binance, OKX	**1.567**	1.610	1.564	supported
Oct 2025 cascade	retail, Binance, Coinbase	**1.663**	1.675	1.667	near-tie

One-step weighted RMSE (lower better); "direction supported" when true clearly beats reversed.

asset (USDC)—and is washed out only when the window is dominated by symmetric custodial settlement. This is exactly the event-conditional reading of A3 advanced above, now supported across three independent episodes spanning 2022–2025.

5.5 Backtests, threats to validity, and summary

As a standard validity check on the risk machinery we backtest a rolling historical VaR on the asset returns. Kupiec unconditional-coverage [26] p-values are 0.24 (BTC) and 0.58 (ETH) at daily resolution; the Christoffersen conditional-coverage test [27] passes for BTC ($p = 0.11$) and flags violation clustering for ETH ($p = 0.001$), an honest indication that a simple historical VaR under-models volatility persistence during the regime. Four threats to validity bound the conclusions. Construct validity is addressed by using multiple stress proxies per entity type and showing that rankings are stable across the Tier-0/Tier-3 graphs; internal validity is threatened chiefly by the off-chain CEX engine (A2), which we cannot observe and whose influence the robustness values bound but cannot eliminate; statistical-conclusion validity motivates the move from the under-powered daily window to the hourly window with 816 predictions; and external validity, limited by a single event, is partially addressed by the agreement with the documented narrative and flagged as the principal direction for future work. Overall, as Table 15 records, the counterfactual-attribution and directional-causality hypotheses (H2a, H2b, H3, H4, H5) are supported, H1 is at best suggestive, and the apparent

Table 15 Hypothesis verdicts at hourly resolution.

ID	Verdict	Evidence
H1	suggestive	observed 1.2529 < 1.2568 (Granger) hourly, but CI spans 0 and not robust to lagged-flow / price-target
H2a	supported	Oct-2025 custodial cascade: true≈reversed (within 1%); connectivity, not direction
H2b	supported	FTX and USDC/SVB: true clearly < reversed; ≈ 9% on leakage-free non-flow target
H3	supported	Spearman vs ΔCoVaR = 0.52; ranking stable (Spearman 0.88 under perturbation)
H4	supported	Shapley-exact decomposition; Binance→Bybit 98% (100% positive under perturbation)
H5	supported	composability edges carry attribution (Ethena→Aave 11.3%, daily)

custodial symmetry (H2a) is itself informative: it tells us that during a deposit-driven cascade the systemic signal lives in connectivity rather than in any single flow direction, whereas the leakage-free non-flow stress (H2b) recovers a clear $\sim 9\%$ true-over-reversed margin. The primary contribution is therefore the verifiable counterfactual attribution layer; one-step predictive improvement is a secondary ablation.

6 Discussion and Conclusion

On real on-chain data spanning three stress episodes (2022–2025), treating the observed transaction graph as a structural causal model yields a systemic-importance ranking that is materially different from price-based measures and that matches the documented event narratives; an exact, order-robust counterfactual loss attribution decomposing realised distress into named inter-entity channels; and a graph that, at the cascade's native hourly resolution, predicts the October-2025 event competitively with—and weakly better than—the estimated-graph state of the art, an advantage we quantify with a paired bootstrap rather than assert. The framework passes placebo, resolution-robustness (including retail-node removal), external-validity, and confounding-sensitivity checks. The most nuanced finding concerns the directional assumption A3. Rather than holding or failing globally, its support is *event- and edge-type-conditional*: edge reversal clearly degrades prediction in the FTX (2022) and USDC/SVB (2023) episodes, whose contagion spreads outward from a failing venue or a depegging asset, but not in the October-2025 cascade, which is dominated by symmetric custodial exchange flows on which direction has little to bite. VEIN should therefore claim directed causal content where the mechanism is genuinely directional—the collateral and composability links emphasised in the recent TradFi–DeFi crosstagion literature [16], and the outward propagation from a stressed hub—and fall back to connectivity-level claims for symmetric settlement flows. This conditional reading, delivered by the falsification machinery rather than assumed, is both sharper and more defensible than a blanket directional claim.

Three limitations bound these conclusions. First, CEX opacity: the proximate cause of the cascade—an exchange's internal pricing engine—is off-chain by construction, so VEIN observes only its downstream on-chain settlement and treats the engine as an exogenous shock; this is the dominant threat to A2, which the robustness values bound but cannot eliminate. Second, event coverage: although we validate across three episodes (2022–2025), broader generalisation across many regimes and a wider entity universe remains future work. Third, resolution recall: residual unlabelled addresses collapse into the retail node, so the inter-entity graph is high-precision but recall-bounded by label coverage—only about 1% of flow volume currently moves directly between two named entities—an improvement frontier the Tier-0/1/2 layers are designed to push. For supervisors, VEIN nonetheless offers two practitioner-facing artefacts that price-based measures cannot: a ranking of which on-chain entities export versus absorb systemic stress, and a counterfactual statement of what a specific entity's distress contributed to others' losses—directly relevant to stress-test attribution and to protocol-level composability-risk assessment, and, because the inputs are public ledger data, auditable and reproducible by any party. The most decisive next steps are intraday (sub-minute) analysis across several stress events, which would give the falsification test the power to localise where, if anywhere, direction is causal; a fuller Tier-1/2 entity-resolution sweep to convert retail-mediated edges into entity-to-entity edges; and an instrumented treatment of the CEX engine (for instance, exploiting exogenous oracle-update timing) to relax A2. In sum, VEIN demonstrates that the blockchain ledger can serve as an *auditable candidate structural skeleton* for interventional and counterfactual systemic-risk attribution, enabling analyses out of reach for price-based methods, while being candid that its empirical edge over the estimated-graph baseline on the primary October-2025 event study—with additional checks on the two earlier stress episodes—is real but resolution-dependent and not decisive. Whether that edge widens with intraday data, fuller entity resolution, and more events is a question for future work to answer, not a result this paper claims.

Data availability. The complete implementation, the real-data evaluation pipeline, and all result artefacts underlying every number, table, and figure in this paper are publicly available at https://github.com/vinhqdang/VEIN-On-Chain-Causal-Risk-Attribution-for-Cryptocurrency-Systemic-Risk. The repository includes the full source code, cached query results, and scripts that reproduce every reported result from the underlying public data. All underlying blockchain and market data are public and were obtained from Dune Analytics (https://dune.com), DefiLlama (https://defillama.com), and Etherscan (https://etherscan.io), as detailed in Section 4; no restricted or proprietary data were used.

Declarations. Ethics approval and consent to participate. Not applicable: this study uses only public, aggregated on-chain and market data and involves no human subjects, human data, or animals.

Consent for publication. Not applicable.

Availability of data and materials. See the Data availability statement above.

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Appendix A Implementation and additional details

The pipeline is implemented in Python. On-chain data are queried via the Dune Analytics SQL API over decoded Ethereum tables, with results cached by query hash for reproducibility and to respect rate limits. Structural equations (2) are fit with ridge regression ($\lambda = 1$); the Monte-Carlo estimator uses $M \in [150, 400]$ residual-bootstrap paths; the Tier-1 classifier is a gradient-boosted tree ensemble and the Tier-2 embedding a 16-dimensional truncated SVD. For the hourly consolidated run the estimation window is 600 hours preceding 10 October 2025, with 48 event hours; severe-distress interventions set s^* to each entity's 95th historical stress percentile and pre-crisis counterfactual levels use the estimation-window median. The VaR/loss level is $q = 0.95$ throughout. Residuals are mean-centred by construction; the block bootstrap resamples residual vectors *jointly* across entities at each step, so contemporaneous cross-entity dependence (common shocks) is preserved rather than destroyed. With $M = 400$ paths the Monte-Carlo standard error of the reported OC-CoVaR quantiles is below 3% of the estimate. Table A1 lists the robustness values for the strongest structural coefficients, computed via (7) from OLS t-statistics on the design of (2). The headline attribution numbers in Table 11 (Section 5.2) are single-entity-removal counterfactuals, computed for every reported upstream/downstream pair; the Shapley decomposition of Figure 7 is computed for one illustrative target (Bybit's event-window distress, over its three candidate on-chain sources) to demonstrate exactness and order-invariance against the single-removal numbers, not recomputed for every channel in the paper—the two are complementary, not interchangeable, and we report results from each under its own name throughout.

Reproducibility. Every result traces to cached artefacts in the project repository. On-chain flows are computed from Dune's `erc20_ethereum.evt_Transfer` joined to `labels.cex_ethereum`; the systemic-core tokens are USDe (`0x4c9e...68b3`), sUSDe (`0x9d39...3497`) and wBETH (`0xa2e3...2ee1`), with USDC (`0xa0b8...eb48`), USDT (`0xdac1...1ec7`) and WETH (`0xc02a...6cc2`) for the event-window enrichment and the 2022–2023 episodes. Event windows are: October cascade 8–14 Oct 2025 (event 10–11 Oct, hourly fit 15 Sep–25 Oct); FTX 1–16 Nov 2022 (event 8–11 Nov); USDC/SVB 1–16 Mar 2023 (event 10–13 Mar). All Monte-Carlo and bootstrap procedures use a fixed seed (7); ridge penalty $\lambda = 1$; resolution threshold $\tau = 0.8$; graph edge threshold US$1M (daily) / US$0.1M (hourly). Each Dune query is persisted by SQL hash so the exact query text and returned rows are recoverable.

Table A1 Robustness values for leading structural coefficients (hourly).

Edge	t**-statistic**	$RV_{q=1}$
retail → Coinbase	7.94	0.226
Coinbase → retail	5.92	0.175
retail → Bullish	5.41	0.160
Binance → Gate.io	5.19	0.154
retail → Bitfinex	4.55	0.137

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